# Datasheets for Datasets

1. A datasheet for a dataset is used to describe the operating characteristics, test results, recommended usage, and other information about the dataset.
2. The authors think datasheets are important because, “datasheets for datasets have the potential to increase transparency and accountability within the ML community, mitigate unwanted societal biases in ML models, facilitate greater reproducibility of ML results, and help researchers and practitioners select more appropriate datasets for their chosen tasks.”
3. The development process for the guidelines set out by the authors are iterative and incremental because they used a repetitive process over the course of two years. The developers leveraged their experiences, created a rough draft and asked lawyers to provide feedback.
4. Two questions they ask are:
    a. “Who created the dataset (for example, which team, research group) and on behalf of which entity (for example, company, institution, organization)?” This question is important to ask so as to avoid any problems on nobody being able to adjust the code if it ever needs to be fixed. Sometimes, code can be hard to read when someone else writes it, especially if it is long and complex. This question will also hold people accountable to try to create readable code, since their name is connected to it.
    b. “For what purpose was the dataset created?” This question is helpful to explain the reasoning behind adding the dataset. The question aims to add an explanation for the need of the dataset being added into the code.

# Debugging Conditionals

1. The UnboundLocalError occurred because there was a possibility that there were no eggs and it was a weekend, but there was no place for the code to go. Thus I added an else statement.
2. The ValueError occurred because there was the possibility for the function to be divided by zero. To solve this, I added an if, else statement to ensure that a > 0. There was also the possibility for there to be an imaginary number, so I added a chain to the if statement to make sure the value exists.
3. The function was printing some incorrect statements such as “wear sunscreen!” when it was too cold for that. I also fixed the function by swapping the and statement for an or statement.

# Write a Homework Problem

## Question:

You are a general of the rebel army in an intergalactic space war against celestial beings. Currently, you are commanding a sneak attack mission on the celestial being king who happens to be the size of a planet. All was going well until right as you were getting close to the king your ships calculations system went down! Now, using the roche limit, you must calculate how close you can get to the king without getting pulled in by his gravitational force! It is important that you get as close as you can while being safe so that you can get the best shot at him!

## Explanation:

The question is meant to encourage the student to solve a problem in a fun and entertaining way. The question has similar themes to a sci-fi war. This theme is just a cover for the actual problem which is finding the Roche limit (similar to the homework problem for this worksheet). In doing so, the solver must create multiple inputs and functions in their code that continuously builds to the point where the answer is

apparent. I believe the most challenging part of the question will be the calculations done by the student as the solutions to steps throughout the problem get used later. It will be difficult for the student to solve the problem if they do not use proper python citation and use poor labels for their values.